
Location, Not Shape: A Controlled Study of Anatomy-Guided Masking for Joint-Embedding Predictive Pretraining on Retinal OCT

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Abstract

1 Masked predictive pretraining asks a model to reconstruct the representation of
2 hidden image regions. Which regions are hidden is a free design choice, and in
3 medical imaging there is an intuitive answer: hide the tissue that carries the di-
4 agnosis. We test that intuition under tightly controlled conditions. Six masking
5 policies are continued from a single shared I-JEPA checkpoint on 3D retinal OCT,
6 differing *only* in how predictor targets are placed, and evaluated with a frozen
7 linear-probe protocol for glaucoma classification (FairVision, $N=3000$). Guid-
8 ance does help: restricting targets to retinal tissue improves AUC by $+0.0120$
9 over unguided masking at the matched epoch (95% CI $[+0.0068, +0.0173]$), and
10 a band located by a first-order intensity statistic improves it by $+0.0109$ at epoch
11 100 ($[+0.0057, +0.0160]$, $p<0.0001$). All six pre-specified contrasts against the
12 null, at all three epochs, exclude zero. But anatomical *precision* does not help.
13 A policy that places 97% of masked patches on tissue, using a trained retinal
14 segmentation model to shape the targets, is *indistinguishable from unguided mask-*
15 *ing* at the matched epoch ($+0.0013$, CI $[-0.0054, +0.0082]$), as is a coverage-
16 constrained policy ($+0.0002$) — while the two policies that merely *place* rect-
17 angles on tissue gain an order of magnitude more. Direct measurement of mask
18 geometry across 600 slices shows the fraction of anatomy hidden is uncorrelated
19 with downstream AUC. The best policy consults no segmentation model at all. A
20 subgroup audit over 19 probes finds the same worst-served group every time; gains
21 reach every group and every disease stage, mild disease most of all ($+0.0137$), but
22 the difficulty ordering is untouched: the severe-to-mild gap stays within 0.0114 of
23 constant across every policy. Because each policy was pretrained once, we report
24 these as single-run observations rather than an established ranking.

25 1 Introduction

26 Self-supervised learning has become the default route to representation quality in medical imaging,
27 where expert annotation is expensive and pathology is rare [Azizi et al., 2021, Zhou et al., 2023, Qiu
28 et al., 2024]. Joint-embedding predictive architectures such as I-JEPA [Assran et al., 2023] learn by
29 predicting the *representation* of masked target blocks from a visible context block, avoiding pixel
30 reconstruction and its bias toward high-frequency detail [He et al., 2022, Xie et al., 2022].

31 I-JEPA samples its targets geometrically: rectangles placed uniformly at random. In natural images
32 this is defensible, since informative content is spread broadly across the frame. In a retinal OCT
33 B-scan it is not obviously so. A large fraction of the image is dark vitreous and background; the

34 diagnostic signal for glaucoma is concentrated in a thin band of retinal tissue. This motivates an
35 intuitive hypothesis, one that several recent methods adopt [Li et al., 2022, 2024, Wang et al., 2025,
36 Shi et al., 2022]:

37 *Direct the predictive objective at clinically meaningful anatomy rather than at*
38 *arbitrary image regions.*

39 We test this hypothesis directly and at several strengths. Rather than proposing one anatomy-aware
40 method and reporting that it beats a baseline, we construct a *ladder* of masking policies of increas-
41 ing anatomical specificity — from unguided rectangles, through rectangles constrained to lie on
42 tissue, to ragged targets whose shape follows the segmented retina — and hold everything else fixed.
43 All arms continue from one shared pretrained checkpoint, use identical optimiser, schedule, and
44 effective batch size, and are evaluated under one frozen-probe protocol.

45 The result contradicts the intuition in an informative way. Guidance helps, but the benefit does
46 not increase with anatomical precision — it disappears. The two policies whose targets track the
47 segmented retina most faithfully are statistically indistinguishable from unguided masking at the
48 matched epoch, while two policies that merely place ordinary rectangles on tissue gain an order
49 of magnitude more. The best performer uses no segmentation model at all, locating a band by a
50 per-column intensity centroid.

51 **Contributions.**

- 52 • A controlled six-arm study isolating masking policy in medical SSL, with a shared ancestor
53 checkpoint and a single evaluation protocol (21 frozen probes).
- 54 • Evidence that guidance helps but anatomical precision does not, including the specific con-
55 trol — rectangles restricted to retinal tissue — that separates “avoid background” from
56 “target anatomy”.
- 57 • Direct measurement of mask geometry for every policy on 600 slices, showing that fraction-
58 of-anatomy-hidden does not order downstream AUC, and an explicit accounting of the axes
59 on which the anatomy arms are confounded.
- 60 • A subgroup audit and a 10^{-5} reproduction of the headline result from public weights on
61 different hardware and precision.

62 **2 Related work**

63 **Masked image modelling and JEPAs.** Masked autoencoders reconstruct pixels [He et al., 2022,
64 Xie et al., 2022]; BEiT [Bao et al., 2022] and iBOT [Zhou et al., 2022] predict discrete tokens;
65 data2vec [Baevski et al., 2022] and I-JEPA [Assran et al., 2023] predict latent representations. Video
66 and 3D extensions follow the same template [Bardes et al., 2024, Chen et al., 2023b, Taleb et al.,
67 2020]. Recent analyses question how much of the benefit derives from the masking distribution
68 itself [Balestriero and LeCun, 2025, He et al., 2025].

69 **Informed masking.** A substantial line of work argues that *what* is masked matters. ADIOS [Shi
70 et al., 2022] learns an adversarial masking model; SemMAE [Li et al., 2022] uses learned se-
71 mantic parts; AttMask [Kakogeorgiou et al., 2022] masks by attention; HardPatches [Wang et al.,
72 2023a] and AutoMAE [Chen et al., 2023a] target difficulty. In the medical domain, AnatoMask [Li
73 et al., 2024], AnatomyMAE [Ceballos Arroyo et al., 2026], MSMAE [Mao et al., 2023], and Va-
74 soMIM [Huang et al., 2026] explicitly steer masking toward anatomy, and Wang et al. [2025] argue
75 directly for masking clinically relevant regions. Our study is a controlled test of that family’s central
76 premise on one task, and finds it does not hold in the direction usually assumed.

77 **Evidence that informed masking is not uniformly better.** The picture is less one-sided than
78 the paragraph above suggests. The original masked-modelling papers already report that random
79 sampling is hard to beat: MAE’s ablation puts random-75 at 73.5 linear against block-75 at 63.9 [He
80 et al., 2022], and SimMIM finds random beats its best block variant [Xie et al., 2022]. Among
81 informed schemes, AttMask sits within 0.2 points of random [Kakogeorgiou et al., 2022], AutoMAE
82 ties MAE under full fine-tuning [Chen et al., 2023a], HPM gains only when randomness is retained

and *loses* when restricted to the hardest patches [Wang et al., 2023b], and attention-guided MAE underperforms MAE in the low-label regime [Sick et al., 2025]. Most directly, SemMAE’s own part-quality ablation shows that *imperfect* semantic parts add nothing over no parts at all [Li et al., 2022], and Guo et al. [2025] reproduce SemMAE below random MAE under a common protocol, with the deficit largest under frozen evaluation — also our protocol. Chen et al. [2023a] name the mechanism a “patch selection dilemma”: a slight foreground bias helps, but raising it too far starves the encoder of the evidence it must predict from. Conversely, the closest positive precedent for our best policy uses no segmentation model either: Lee et al. [2025] select foreground from normalised intensity alone and gain across five 3D medical benchmarks. Appendix G tabulates these.

We therefore do *not* claim to overturn the informed-masking literature. That random masking is a strong baseline, and semantic guidance wins only sometimes, is known and contested. What we add is a controlled measurement of where a *trained medical segmentation model* falls on that spectrum when everything else is held fixed, in a setting — 3D retinal OCT, joint-embedding prediction, frozen probe — where we could find no prior controlled comparison.

Retinal foundation models and OCT. RETFound [Zhou et al., 2023], VisionFM [Qiu et al., 2024], URFound [Yu et al., 2024], and OCTCube [Liu et al., 2026] pretrain on large retinal corpora; OCT-specific masked modelling is explored by Pissas et al. [2024]. MIRAGE [Morano et al., 2025] provides multimodal OCT segmentation and supplies the anatomical guidance used by several of our arms. Deep glaucoma detection from OCT is established [Maetschke et al., 2019, Ran et al., 2019, Noury et al., 2022].

Fairness in medical imaging. Performance disparities across demographic groups are widely documented [Seyyed-Kalantari et al., 2021, Larrazabal et al., 2020, Gichoya et al., 2022]. FairVision [Luo et al., 2024a] and Harvard-GF [Luo et al., 2024b] provide OCT data with demographic attributes; FairMedFM [Jin et al., 2024] and Shi et al. [2025] benchmark fairness for medical foundation models. We report a subgroup analysis as a secondary finding.

3 Method: a ladder of masking policies

3.1 Background

I-JEPA partitions the token grid of an image into a context block and M target blocks. A context encoder f_θ embeds the visible tokens; a predictor g_ϕ maps that embedding plus positional queries to predicted representations of the target tokens; an exponential-moving-average target encoder f_θ supplies the regression targets. With a 16×16 token grid the design choice we vary is which of the 256 cells become targets.

3.2 The six policies

All policies produce $M=4$ target blocks and one context block under identical scale and aspect-ratio ranges. They differ only in placement and shape.

RANDOM (null). Stock I-JEPA multiblock. Four rectangles placed uniformly at random anywhere in the grid. No image content is consulted.

ENVELOPE (random-within-retina). Identical rectangles — same shape, size and count as RANDOM — but rejection-sampled onto the retinal envelope predicted by MIRAGE [Morano et al., 2025]. Only *location* changes. This is the control that separates “stop wasting targets on background” from “target anatomy specifically”.

INTENSITY (segmentation-free). A band whose vertical position is located per slice by a per-column intensity-weighted row centroid, smoothed across columns. It adapts to the retina’s position and curvature using a first-order intensity statistic, and consults **no segmentation model**. It uses no ground-truth annotation of any kind. (Our released code and checkpoints label this policy `oracle`, a historical name from when it was intended as an upper bound. We rename it here because it is neither an oracle nor an upper bound: it reads only the input image.)

Six masking policies on one B-scan, drawn with the production samplers.
Anatomical specificity increases left to right, top to bottom.

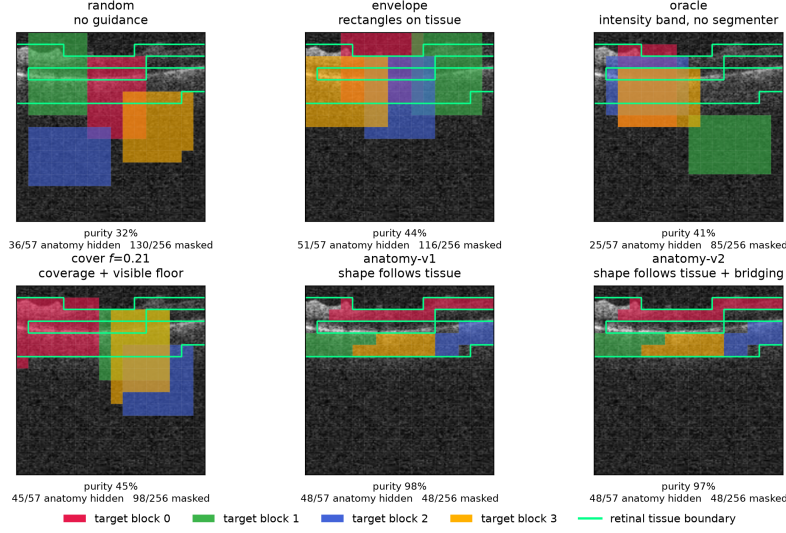


Figure 1: The six masking policies on one identical B-scan, drawn with the production samplers. Each predictor target block has its own colour; the green contour is the anatomy reference. Anatomical specificity increases along the ladder: ENVELOPE and COVER place rectangles on tissue, while the anatomy arms grow targets that trace the retina almost exactly. Downstream AUC does not increase with that specificity (Table 1). Figure 4 repeats this over five B-scans spanning the volume.

- 131 **ANATOMY-V1 / ANATOMY-V2.** Ragged connected components grown on the MIRAGE score map,
132 so target *shape* follows tissue rather than being rectangular. v2 additionally bridges diago-
133 nal adjacencies.
- 134 **COVER f .** Targets chosen greedily to cover the anatomy subject to a hard floor leaving a fraction f
135 of tissue visible in the context. We pretrain $f=0.21$.

136 Figure 1 shows all six on the same B-scans.

137 3.3 Hypotheses

138 The ladder yields three testable statements:

- 139 **H1** *Avoiding background helps.* ENVELOPE > RANDOM.
- 140 **H2** *Anatomical precision helps further.* anatomy arms > ENVELOPE.
- 141 **H3** *Any benefit is due to anatomical targeting, not to incidental changes in mask area, target count,*
142 *or context size.*

143 4 Experimental setup

144 **Data.** FairVision [Luo et al., 2024a] glaucoma OCT volumes. Each volume is a 3D scan resampled
145 to 100 B-scans of 256×256 after transform. The label is a binary glaucoma diagnosis. Splits are
146 fixed and patient-disjoint as released; downstream evaluation uses a held-out test split of $N=3000$
147 volumes (1466 positive / 1534 negative), seed 42, fixed loader order. No test volume is seen during
148 pretraining, probe fitting, or model selection: the probe head is selected on a separate validation
149 split. Labels are byte-identical across arms, so any two arms are paired-comparable on the same
150 cases. The dataset carries self-reported demographic attributes, which we use only for the subgroup
151 analysis in Section 5.3 and never as model inputs.

Table 1: Frozen MeanPool probe, FairVision test ($N=3000$, 1466 positive). All arms continue from one epoch-25 ancestor (AUC 0.8487). Δ is versus RANDOM at the *matched* epoch. The upper block is precision-matched (fp16 throughout) and carries every headline claim. The lower block is fp32 and its Δ therefore crosses the precision boundary; those rows are marked $\ddagger$ and are excluded from headline claims.

policy	guidance signal	epoch 50		epoch 75		epoch 100	
		AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	none (null)	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	segmenter (location only)	0.8761	+0.0120	0.8803	+0.0080	0.8807	+0.0062
INTENSITY	intensity centroid, no segmenter	0.8740	+0.0099	0.8836	+0.0113	0.8855	+0.0109
ANATOMY-V2	segmenter (target shape)	0.8654	+0.0013 $\ddagger$	—	—	—	—
COVER $f=0.21$	segmenter (coverage)	0.8643	+0.0002 $\ddagger$	—	—	<u>pending†</u>	<u>pending†</u>

Pretraining. Patch-level I-JEPA, ViT-Base, 256×256 crops, patch size 16, predictor depth 6, EMA target encoder. **Every arm resumes from the same epoch-25 checkpoint** and continues with identical optimiser, learning-rate and weight-decay schedules, and effective batch size (512 for all arms). Guidance is ramped linearly from epoch 25 to 30 and held at full strength thereafter. The *only* difference between arms is the mask sampler.

Arms differ in how far they were carried, and we do not pretend otherwise. RANDOM, INTENSITY and ENVELOPE reach epoch 100. ANATOMY-V2 has valid probes to epoch 50 (later checkpoints follow a target-precision splice and are excluded, Appendix D). ANATOMY-V1 has only epoch 30. COVER $f=0.21$ was deliberately halted at epoch 73. Epoch 50 is therefore the only epoch at which five policies are simultaneously available, and it carries every cross-policy comparison that involves the anatomy or coverage arms.

Evaluation. The encoder is frozen. Per-slice features are mean-pooled over the volume, followed by LayerNorm and a linear head. The head is selected by validation AUC and reported on the held-out test split. We report paired bootstrap confidence intervals over test volumes (10,000 resamples, stratified by class, the same resampled index set applied to every arm) and DeLong tests for correlated ROC curves [DeLong et al., 1988], both of which are valid here because all arms are evaluated on the same cases in the same order. Each test volume is a distinct subject, so case-level resampling is already subject-level and no cluster correction is required [Obuchowski, 1997, Rutter, 2000]. We follow Maier-Hein et al. [2024] in reporting the operating-point metrics alongside AUC rather than AUC alone (Appendix A).

Numerical precision, and how we handle it. The probe harness defaults to mixed precision. The RANDOM, INTENSITY and ENVELOPE long-horizon probes were fitted under that default (fp16); the ANATOMY-V1, ANATOMY-V2, COVER and ancestor probes were fitted with autocast explicitly disabled (fp32). We therefore do *not* claim a single protocol across every probe in the study. Instead we partition the comparisons: all headline contrasts in Table 1 are drawn from arms sharing both epoch *and* precision, and any contrast that would cross the precision boundary is marked as such and excluded from headline claims. Section 5.4 reports a direct measurement of how large the precision effect actually is.

5 Results

What these numbers can and cannot establish. Each policy was pretrained exactly once, so policy is perfectly confounded with one stochastic optimisation path after the fork. The intervals below quantify sampling error over test subjects for a *fixed* pair of score vectors, not seed-to-seed variance of pretraining. Read every comparison here as a controlled single-run observation, not an established ranking over retrainings (Section 6).

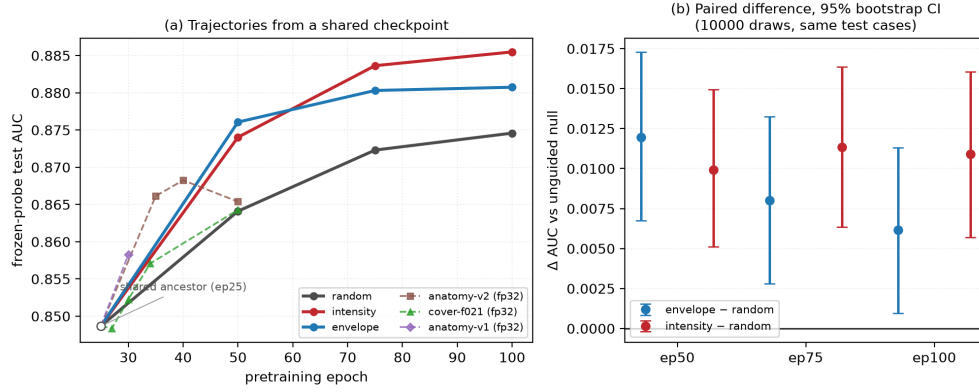


Figure 2: **(a)** Frozen-probe test AUC against pretraining epoch. All arms fork from the same epoch-25 checkpoint (open circle) and differ only in masking policy. Solid lines are the precision-matched fp16 family; dashed lines are fp32 arms and must not be compared vertically against the solid lines. **(b)** The actual inference: paired differences against the unguided null on identical test cases, with 10,000-resample bootstrap intervals. All six intervals exclude zero. We plot paired differences rather than per-arm error bars because marginal intervals carry between-case variance that cancels in a paired comparison, and would understate the evidence.

186 5.1 H1 holds, H2 fails

187 Table 1 gives the main comparison, Table 7 in Appendix H the full paired inference for all
 188 nine precision-matched contrasts, and Figure 2 the trajectories. All six contrasts against the
 189 null survive Benjamini–Hochberg correction over that family of nine ($q \leq 0.0299$ throughout).
 190 **H1 holds:** constraining the same rectangles to lie on retinal tissue (ENVELOPE) beats unguided
 191 masking by $+0.0120$ at epoch 50 (CI $[+0.0068, +0.0173]$) and by $+0.0062$ at epoch 100 (CI
 192 $[+0.0010, +0.0113]$). Avoiding background is worth something, at every epoch we measured, with
 193 an interval that excludes zero.

194 **H2 fails.** The arms that use the segmentation model most aggressively — shaping targets to trace the
 195 retina, or maximising anatomy coverage under a floor — do not improve on the null. At epoch 50,
 196 the only epoch at which all five arms have a probe, ANATOMY-V2 gains $+0.0013$ over RANDOM and
 197 COVER gains $+0.0002$, against $+0.0120$ for ENVELOPE and $+0.0099$ for INTENSITY. The policies
 198 that merely *place* rectangles on tissue gain roughly an order of magnitude more than the policies
 199 that shape targets to match it. Increasing anatomical fidelity past “rectangles on tissue” does not add
 200 benefit.

201 We are careful not to overstate this. ANATOMY-V2 is not *worse* than the null at the matched epoch;
 202 it is statistically indistinguishable from it (CI $[-0.0054, +0.0082]$, $p = 0.7101$), as is COVER (CI
 203 $[-0.0049, +0.0053]$, $p = 0.9440$). Both of those intervals cross the fp16/fp32 boundary and so are
 204 quoted with the ‡ caveat of Table 1; Section 5.4 shows the precision effect is roughly two orders
 205 of magnitude smaller than the interval width, so the conclusion does not turn on it. What the data
 206 rule out is not that these policies are harmful, but the expected *monotone improvement with anatomical precision*: the intervals are centred near zero and comfortably contain it, while the two cruder
 207 policies produce intervals that exclude it.
 208

209 The strongest policy is the one that uses no segmentation model. INTENSITY gives $+0.0109$ over
 210 the null at epoch 100 (95% CI $[+0.0057, +0.0160]$, $p < 0.0001$, $q = < 0.0001$), with a margin that is
 211 stable across training ($+0.0099, +0.0113, +0.0109$). It exceeds the segmenter-guided ENVELOPE
 212 arm at epoch 100 by $+0.0047$ (CI $[+0.0004, +0.0091]$, $p = 0.0294$), though the two are indistin-
 213 guishable at epochs 50 and 75. Notably, ENVELOPE’s margin over the null *decays* with training
 214 ($+0.0120 \rightarrow +0.0080 \rightarrow +0.0062$) while INTENSITY’s does not.

Table 2: Measured mask geometry, 600 held-out slices, production samplers. “purity” is the fraction of masked cells lying on tissue; “loss slots” is the number of predictor targets contributing to the loss. All values measured. AUC is quoted at the *matched* epoch 50, the only epoch at which all five policies have a probe, so that the geometry-to-AUC association is not read across mixed epochs. The two fp32 arms are marked ‡ as in Table 1.

policy	anatomy hidden	purity	mask ratio	context kept	loss slots	AUC @ep50
RANDOM	52.2%	31.6%	43.7%	42.1%	157.7	0.8641
INTENSITY	62.2%	41.1%	40.0%	45.6%	158.4	0.8740
ENVELOPE	76.9%	43.5%	46.4%	40.5%	159.9	0.8761
COVER	74.1%	45.3%	43.3%	43.5%	160.0	0.8643 [‡]
ANATOMY-V2	80.3%	97.3%	21.4%	67.9%	64.0	0.8654 [‡]

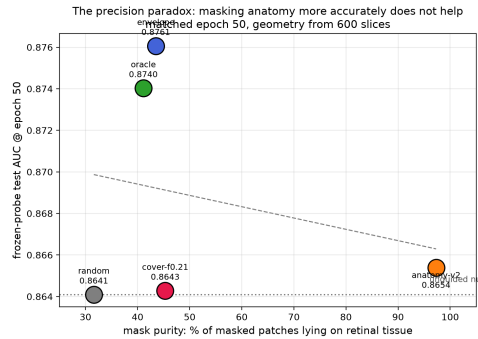


Figure 3: Downstream AUC at matched epoch 50 against mask purity, the fraction of masked patches lying on retinal tissue. The policy whose masks are 97% on-tissue does not separate from the null, while policies at 41–44% purity gain the most. The ordering does not support the intuition that more accurate anatomical masking yields better representations. With four to five arms this is descriptive, not inferential; the same result arranged as a specificity ladder is in Appendix I.

215 5.2 H3: it is not anatomical targeting

216 To test H3 we ran each production sampler on 600 held-out slices and measured what it actually
 217 does (Table 2, Figure 3). Three observations follow.

218 First, **anatomical targeting does not order the results**. Over the four rectangle arms, the Spear-
 219 man correlation between fraction-of-anatomy-hidden and AUC is exactly 0.00: knowing how much
 220 anatomy a policy hides tells you nothing about its downstream AUC. Mask *purity* — the share of
 221 masked patches lying on tissue — trends negatively (−0.40). Neither is significant at $n=4$ arms, and
 222 we present them as descriptive rather than inferential; but neither shows the positive relationship the
 223 anatomy-guidance hypothesis predicts. The anatomy arm places 97.3% of masked cells on tissue
 224 and does not separate from the null; INTENSITY places 41.1% on tissue and gives the largest gain in
 225 the study.

226 Second, **the anatomy arms are confounded on several axes simultaneously**, and we state this
 227 rather than attributing their deficit to shape. Because their targets are compact tissue blobs rather
 228 than large rectangles, they hide only 21.4% of the grid against 40–46% for the rectangle arms, leave
 229 67.9% of tokens visible as context against 40–46%, and supply 64 predictor loss slots against 158–
 230 160. Any of these could account for their deficit without invoking target shape at all. We therefore
 231 do *not* claim that irregular target shape is harmful; that comparison is not identified by this design.
 232 Figure 12 makes the four-way divergence explicit.

233 Third, INTENSITY is the most *spatially consistent* policy we measured. Its per-cell hit map is the
 234 most concentrated of any arm (Gini 0.400 versus 0.316–0.338 for the other rectangle arms): it
 235 repeatedly asks the encoder to represent the same region of the image. This suggests a mechanism
 236 other than anatomical correctness — consistency of the predictive task — which we return to in
 237 Section 6.

238 5.3 Subgroup analysis

239 Joining saved test predictions to the released FairVision metadata in deterministic loader order (the
240 join is validated by exact label reconstruction: the label vector rebuilt from the sorted file order
241 matches the stored labels for all 3000 cases in every probe), we audit seven stratifications across 19
242 probes spanning aggregate AUC 0.8483 to 0.8855. Probes that the exclusion rules of Appendix D
243 remove are removed here too.

244 **The ordering of subgroups never changes.** The worst-performing group is the same in every
245 one of the 19 probes for sex (female), race (black), ethnicity (hispanic) and disease severity (mild
246 $(-6, -2]$). No masking policy reorders them. The probes share a test set, an epoch-25 ancestor
247 and a probe seed, and several are different epochs of one branch, so they are *not* independent; we
248 report this as a descriptive regularity and attach no p -value. What it shows is that the disparity is not
249 introduced by any policy we tried.

250 **Every group improves; no gap reliably moves.** A widening max-min gap is easy to misread as
251 harm. It is not. At epoch 100 the strongest policy improves *every* race stratum over the null — black
252 $0.8325 \rightarrow 0.8472$ (+0.0147, a larger gain than white’s $0.8741 \rightarrow 0.8853$) — and the max-min gap
253 widens only because the already-best asian subgroup gains most. Nor is there a defensible gap *trend*:
254 the race gap correlates with aggregate AUC at $\rho = +0.511$ ($p = 0.0255$) across checkpoints, but
255 this fails Benjamini-Hochberg correction over the seven attributes tested ($q = 0.0595$) and vanishes
256 when pseudo-replication is removed by averaging within each of the 7 branches ($\rho = +0.429$,
257 $p = 0.3374$). The only gaps that survive both are sex and severity, and both *narrow*. We do not
258 claim that better masking harms any group, and we do not claim it helps equity either.

259 **Early disease improves, but stays hard.** Scoring each severity stratum against the shared pool
260 of all negatives, the strongest policy improves mild disease from 0.8164 to 0.8301 (+0.0137 at
261 matched epoch 100), moderate by +0.0102 and severe by +0.0063: mild disease improves *most*.
262 What does not move is the difficulty ordering. The severe-to-mild gap is 0.1361 for the null and
263 0.1286 for the best policy, and stays within 0.0114 of constant across all 19 probes while aggregate
264 AUC spans 0.8483–0.8855. Target placement lifts the whole curve; it does not change its shape.
265 Appendix C gives all seven stratifications. These are secondary, descriptive analyses; no policy here
266 was designed as a fairness intervention and none should be presented as one.

267 5.4 Numerical precision is not the effect

268 Because the long-horizon arms and the anatomy arms were probed at different autocast settings
269 (Section 4), we measured the precision effect rather than assuming it away. Re-encoding the epoch-
270 100 INTENSITY test split at fp32 reproduces the reported AUC to 9.8×10^{-6} , and a full re-fit of the
271 entire probe pipeline at fp32 shifts every long-horizon arm by less than 2×10^{-4} . Both are two to
272 three orders of magnitude below the effects in Table 1. Appendix I gives the numbers.

273 5.5 Reproducibility

274 The epoch-100 INTENSITY encoder and its trained probe head will be released, together with the
275 stored per-case predictions for every probe in this paper. Links are withheld here for anonymity.
276 Every number in this paper is emitted by a single script from those stored predictions, so tables,
277 figures and prose cannot disagree; that script is released with the code.

278 6 Discussion and limitations

279 **What the evidence supports.** Guidance helps: restricting targets to tissue is worth +0.0120 at the
280 matched epoch, and a consistent intensity-located band is worth +0.0109 at epoch 100, with paired
281 intervals excluding zero at every epoch measured. A trained segmentation model is not required to
282 obtain this benefit, and using one more aggressively — to shape targets or to maximise anatomy
283 coverage — does not add to it. The practical reading, bounded by the caveats below, is that for *this*
284 task the segmentation stage bought us nothing we could measure, and a one-line intensity statistic
285 recovered the whole benefit at a fraction of the engineering cost. We would not generalise that to

tasks where anatomy is the output — layer segmentation, lesion localisation, progression — where segmentation-aware pretraining may well matter.

What it does not support. We cannot claim anatomy-shaped masking is *harmful*: at the matched epoch it sits $+0.0013$ from the null with an interval $([-0.0054, +0.0082])$ that comfortably contains zero, so it is indistinguishable rather than worse. Nor can we claim target *shape* is the operative variable, for the reasons in Table 2. Nor can we claim the mechanism behind INTENSITY’s advantage. Two explanations remain live: consistency of location (the encoder is repeatedly forced to represent the same region), and masking ratio or task difficulty. Distinguishing them requires an arm that randomises the band’s vertical position while holding area and context fixed, which we did not run.

One pretraining run per policy. The paired bootstrap quantifies sampling error on a fixed test set; it does not quantify seed-to-seed variance of pretraining. With $n=1$ continuation per arm the *ranking* is not statistically established even where individual pairwise differences are significant on this test set. Bouthillier et al. [2021] show that single-seed comparisons routinely mistake optimisation noise for method effects. We can bound one component: with five probe seeds on two fixed encoders the probe-seed standard deviation is 0.0003 and 0.0018, and the arms stay fully separated across all five, so probe noise is an order of magnitude below the effects in Table 1. Those are *technical* replicates and do not estimate pretraining noise, which would need at least three paired continuations from the shared checkpoint with the continuation as the unit of replication. That is the single most valuable follow-up and we do not claim its result in advance.

Scope, coverage and an incomplete arm. All results are glaucoma classification from FairVision OCT; generality to other modalities, diseases, or segmentation-quality regimes is untested, and the anatomy arms depend on MIRAGE’s prediction quality on this data, so a better segmenter might change the conclusion. Only $f=0.21$ was pretrained for COVER, so we report no dose-response over the visible-tissue floor and make no claim that any floor is optimal. That arm also has probes only at epochs 27, 30, 34, 50 and 73: it was deliberately halted before epoch 100, its later probes are marked pending rather than reported, and we draw no conclusion about its endpoint. Its epoch-50 value is included because it is a matched-epoch comparison against every other arm. Finally, one dataset and one test split were reused throughout this programme’s history, and policies, checkpoints and analyses were chosen after repeated inspection of that split; the intervals we report are therefore best read as descriptive rather than as confirmatory inference.

Broader impact and ethics. This study uses a public, de-identified retinal dataset under its release terms and involves no new data collection or human subjects. No model reported here is clinically validated or intended for deployment. Appendix F gives the full statement.

7 Conclusion

We tested the intuition that masked predictive pretraining in medical imaging should target clinically meaningful anatomy. Under matched compute, schedule, and a shared ancestor checkpoint, guidance helped — but anatomical precision did not. Restricting targets to retinal tissue improved AUC over unguided masking by $+0.0120$ at the matched epoch, while shaping targets to trace the segmented retina, or maximising anatomy coverage, left performance indistinguishable from not guiding at all; and the best policy consulted no segmentation model. Mask geometry is uncorrelated with how much anatomy is hidden, and a subgroup audit over 19 probes shows the gains do not reach the groups or the disease stage that need them most. We suggest the operative variable is the consistency of the predictive task rather than its anatomical correctness, and will release weights and the reproduction protocol.

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539 A All frozen probes

540 Table 3 lists every frozen MeanPool probe in the study.

Table 3: All frozen probes. Protocol identical throughout: FairVision test $N=3000$, seed 42, frozen encoder, MeanPool + LayerNorm + Linear.

policy	epoch	test AUC	target precision
ancestor (RANDOM @ep25)	25	0.848680	fp32
RANDOM	50	0.864097	fp32
RANDOM	75	0.872302	fp32
RANDOM	100	0.874581	fp32
INTENSITY	50	0.874030	fp32
INTENSITY	75	0.883636	fp32
INTENSITY	100	0.885485	fp32
ENVELOPE	30	0.853917	fp32
ENVELOPE	50	0.876064	fp32
ENVELOPE	75	0.880307	fp32
ENVELOPE	100	0.880743	fp32
ANATOMY-V1	30	0.858274	fp32
ANATOMY-V2	35	0.866129	fp32
ANATOMY-V2	40	0.868251	fp32
ANATOMY-V2	50	0.865386	fp32
ANATOMY-V2	75	0.862492	fp16 (confounded)
ANATOMY-V2	92	0.860364	fp16 (confounded)
COVER $f=.21$	27	0.848347	fp32
COVER $f=.21$	30	0.852249	fp32
COVER $f=.21$	34	0.857083	fp32
COVER $f=.21$	50	0.864281	fp32

541 B Masking policies across the volume

542 Figure 1 shows one B-scan for legibility. Figure 4 repeats the same rendering on five B-scans
543 sampled across the depth of a volume, so that the stability of each policy under changing retinal
544 geometry can be inspected. The samplers, the colour assignment, and the anatomy contour are
545 identical to Figure 1.

546 C Subgroup and severity analysis

547 All numbers below come from saved test predictions joined to FairVision’s released metadata in
548 deterministic loader order; the join is validated by exact label reconstruction (3000/3000 labels
549 recovered in every probe). The 19 probes used here are exactly those the inventory marks valid: the
550 four retracted coverage probes of Appendix D and the two precision-spliced ANATOMY-V2 probes
551 are excluded here as everywhere else in the paper.

552 **Why we report two sample sizes.** The 19 probes span only 7 pretraining branches, because sev-
553 eral are different epochs of one training run. Probes within a branch share an encoder lineage, a
554 seed and the test split, so treating them as 19 independent observations is pseudo-replication. Ta-
555 ble 4 therefore reports each trend twice: once per checkpoint, and once after averaging within each
556 branch. Where the two disagree, the branch-level figure is the one to believe.

557 **Race.** The black subgroup has the lowest AUC in every one of the 19 probes ($n=431$ black, $n=251$
558 asian, $n=2318$ white). At matched epoch 100 the black subgroup improves from 0.8325 under
559 RANDOM to 0.8472 under INTENSITY (+0.0147), a larger absolute gain than the white subgroup’s
560 ($0.8741 \rightarrow 0.8853$). The max–min gap nonetheless widens slightly, because the asian subgroup —
561 already the highest — gains most ($0.9033 \rightarrow 0.9189$). A widening max–min gap with universally
562 rising subgroup AUCs is not evidence of harm, and we do not present it as such. The checkpoint-
563 level correlation between aggregate AUC and racial gap is $\rho = +0.511$ ($p=0.0255$), which fails BH
564 correction ($q=0.0595$) and vanishes at branch level ($\rho = +0.429$, $p=0.3374$). We therefore explicitly
565 do *not* claim that more accurate models are less fair.

566 **Disease severity.** FairVision’s label is defined by visual-field mean deviation (md): every test
567 volume with $md \leq -2$ is positive. Severity therefore cannot be used as an ordinary subgroup axis,

Table 4: Subgroup gap trends. “gap range” is the max–min subgroup AUC gap observed across the 19 valid probes. q is Benjamini–Hochberg across the seven attributes tested, which is necessary because seven trends were examined. Only the sex gap and the severity gap survive correction, and both *narrow* as aggregate AUC rises.

attribute	worst group	gap range	per checkpoint ($n=19$)			per branch ($n=7$)	
			ρ	p	q	ρ	p
sex	female (19/19)	0.0272–0.0396	-0.714	0.0006	0.0042	-0.857	0.0137
race	black (19/19)	0.0475–0.0935	+0.511	0.0255	0.0595	+0.429	0.3374
ethnicity	hispanic (19/19)	0.0223–0.0661	+0.351	0.1408	0.1971	+0.464	0.2939
language	other languages (15/19)	0.0518–0.0805	+0.275	0.2537	0.2537	+0.786	0.0362
marital status	single (19/19)	0.0364–0.0654	-0.444	0.0570	0.0997	-0.714	0.0713
age	<60 (18/19)	0.0487–0.0704	+0.304	0.2065	0.2409	+0.357	0.4316
disease severity	mild (−6, −2] (19/19)	0.1286–0.1400	-0.521	0.0222	0.0595	-0.857	0.0137

because within-bin AUC is undefined — a trap that silently yields nothing under naive stratification. Instead each positive stratum is scored against the shared pool of all 1534 negatives.

Table 5: Severity-stratified AUC at matched epoch 100. Each positive stratum is scored against the shared pool of all 1534 negatives. Every stratum improves, and mild disease improves most.

stratum	n_+	RANDOM	ENVELOPE	INTENSITY	Δ
severe ($md \leq -12$)	334	0.9525	0.9559	0.9588	+0.0063
moderate ($-12 < md \leq -6$)	460	0.9030	0.9103	0.9131	+0.0102
mild ($-6 < md \leq -2$)	672	0.8164	0.8232	0.8301	+0.0137

The severe-to-mild gap is 0.1361 for the null and 0.1286 for the strongest policy, and stays within 0.0114 (0.1286–0.1400) across all 19 probes while aggregate AUC spans 0.8483–0.8855. Target placement raises the whole severity curve — mild disease most of all — without altering the ordering that makes early disease hard. We note that mild functional loss is harder to separate from normal OCT by construction, so this ordering is expected clinically; what the data add is that no masking policy we tested changes it.

Limitations. One dataset, one test split, reused throughout this programme’s history. Because policies, checkpoints and analyses were chosen after repeated inspection of that split, the p -values here should be read as descriptive rather than as confirmatory inference. Race and ethnicity are self-reported categories inherited from FairVision’s coding; at these subgroup sizes we cannot speak to intersectional groups. All figures are frozen mean-pool linear probes with AUC as the only metric; we report no subgroup calibration, no sensitivity at a fixed specificity, and no threshold-transfer analysis, so this audit should not be read as a complete fairness evaluation. Disparities could differ under fine-tuning or a stronger head.

D Excluded runs

Four probes of an earlier coverage arm are excluded from all analysis. They were trained with half-precision EMA targets and a different encoder-truncation policy from the arms they would be compared against, so their deficit is not attributable to masking. We report their existence for completeness and do not use them to support any claim. Similarly, the ANATOMY-V2 probes at epochs 75 and 92 follow a change of target precision partway through that arm’s training; they are listed in Table 3 but excluded from the matched-epoch comparisons in Table 1.

E Occlusion attribution: where the classifier looks

The analyses in this appendix were run on the three *fine-tuned* probes (mean-pool, cross-attention pool, and a depth-1 attentive probe) that tie at test AUC ≈ 0.887 , not on the frozen probes used for the arm comparison in the main text. They characterise what the downstream classifier uses, and they are reported here because two of the findings are cautionary rather than positive.

596 **Method.** Attribution is by *occlusion*, which is architecture-agnostic and causal. Gradient×input
597 is not used because two of the three probes are non-linear in the slice-token matrix, where gradient
598 attribution is misleading. For slice-level attribution each slice token is zeroed in turn and Δlogit
599 recorded; for window-level, seven consecutive slices; for patch-level, leave-one-out over the 256
600 patch tokens of a target slice.

601 **A bimodal attribution curve that is an artefact, not anatomy.** The population-averaged curve
602 peaks at native slice ≈ 63 and ≈ 137 with a dip near 95. The tempting reading — that the model has
603 discovered the superior and inferior optic disc rim — is **wrong**, and we report it because it is exactly
604 the kind of claim a plausible-looking attribution figure invites. Three tests reject it. First, if the
605 two peaks were bilateral anatomy then within a diseased eye they should co-occur, giving positive
606 per-volume correlation between them; the observed correlation is slightly *negative* (-0.22 , -0.07 ,
607 -0.14 across the three probes). Second, clustering the per-volume curves into two groups yields
608 clusters that are near-perfect *mirror images* along the slice axis ($\text{corr}(c_1, \text{flip}(c_2)) = 0.971$ and
609 0.988 for the two well-structured probes, against raw correlations of -0.124 and -0.478). Third,
610 using that cluster label as pseudo-laterality and flipping one cluster collapses the bimodality asym-
611 metrically. The signature is OD/OS axial storage: each eye carries its dominant signal on one side
612 of the disc, right and left eyes are stored with opposite axial orientation, and population averaging
613 manufactures an illusion of symmetry.

614 **Errors are weaker signal, not different anatomy.** Stratifying by outcome, false-negative curves
615 are scaled-down true-positive curves with the same shape, and false positives likewise mirror true
616 negatives. Attribution structure is also near-invariant to prediction confidence ($|r| \leq 0.25$ between
617 peak contribution magnitude and $|\text{logit}|$). The error rate at threshold 0.5 therefore reflects signal-
618 strength saturation on hard cases, not the model attending to the wrong structures.

619 **Why this is in a masking paper.** Two of these results bear directly on the main claim. The classi-
620 fier’s evidence is spread across the middle two-thirds of the volume and across most patches within
621 a slice, rather than concentrated on a compact anatomical target. A masking policy that sharpens
622 targets onto a precise anatomical boundary is therefore optimising for a spatial prior the downstream
623 task does not appear to need, which is consistent with Section 5.2 finding no relationship between
624 anatomical specificity and downstream AUC. The OD/OS result is a methodological caution of the
625 same family as our precision and epoch-matching checks: a figure that looks anatomically meaning-
626 ful can be an artefact of data storage, and we would rather report that than let it stand.

627 F Broader impact and ethics

628 This study uses a public, de-identified retinal dataset (FairVision) under its release terms and involves
629 no new data collection and no human subjects. No model reported here is clinically validated or
630 intended for deployment: the frozen-probe AUCs are research measurements on one dataset and one
631 task, and are not evidence of clinical utility.

632 The subgroup analysis in Section 5.3 is a caution, not a contribution. Every masking policy we
633 tested leaves the same groups worst-served, and the best-performing policy does not close any gap.
634 We would regard deployment of any encoder reported here, without a dedicated and prospective
635 fairness evaluation on the target population, as inappropriate. We also stress that the audit is AUC-
636 only: it contains no subgroup calibration, no sensitivity at a fixed clinical specificity, no predictive-
637 value analysis and no intersectional breakdown, so it should not be mistaken for a complete equity
638 assessment.

639 Releasing pretrained weights carries the usual dual-use consideration that they may be applied to
640 populations unlike the training cohort. We document the cohort composition (Appendix C) so that
641 such a mismatch is detectable by anyone reusing the weights. The demographic attributes are self-
642 reported categories inherited from the dataset’s coding; they are used only for the subgroup audit
643 and are never model inputs.

G Published ablations on informed versus random masking

Our result should be read against the following published comparisons, which we collected while positioning this work. The pattern is that random masking is a strong baseline and informed masking wins only under some objectives and protocols; the deficit is largest under frozen evaluation.

Table 6: Published ablations comparing a random or simple baseline against an informed or structured masking scheme. Numbers are as reported by the cited papers; metrics differ across rows and are not comparable between rows.

study	baseline	informed variant	interpretation
MAE [He et al., 2022]	random-75: 84.9 FT / 73.5 lin	block-75: 82.8 / 63.9; grid-75: 84.0 / 66.0	random decisively better
SimMIM [Xie et al., 2022]	83.0 top-1	best block 82.7; square 82.6	random modestly better
SemMAE [Li et al., 2022]	random 66.8 lin	within-part 66.5; whole-part 52.9; adaptive 68.7	naive semantics hurt; scheduled mantics help
SemMAE part quality [Li et al., 2022]	no parts 63.7	raw parts 63.6; refined 65.0	an imperfect semantic model a nothing
AutoMAE [Chen et al., 2023a]	MAE 83.26 FT	AutoMAE 83.32	effectively tied
HPM [Wang et al., 2023b]	random 82.49	moderate 82.95; hardest-only 81.40	benefit only with retained randomness
AttMask [Kakogeorgiou et al., 2022]	random 43.4 k -NN	high 43.5; hint 43.6	nearly indistinguishable
AttG-MAE [Sick et al., 2025]	MAE 78.5 @10% labels	guided 78.1	guidance loses in this protocol
SSiT [Huang et al., 2024]	no saliency 79.48 κ	25% mask 77.53; 50% 79.97	dataset-dependent, non-monotonic

Read together, these say that the *direction* of our finding is not new. What we add is a controlled measurement in a setting where the guidance comes from a trained medical segmentation model rather than from attention or difficulty, and where the comparison is run from a shared checkpoint with the masking policy as the only difference.

H Full paired-contrast table

I Full fp32 re-probe

The RANDOM, INTENSITY and ENVELOPE long-horizon probes were originally fitted under the harness default, which enables autocast. To remove any doubt that the headline effect is a precision artifact, we re-ran the complete probe pipeline for those arms with autocast disabled, so that feature extraction, head optimisation and test evaluation are all fp32, under the exact protocol already used by the ANATOMY and COVER arms. The encoder checkpoint is SHA-256 hashed before and after each run and verified unchanged, so no result here can be contaminated by an accidental encoder update.

[fp32 re-probe table pending[†]](#)

J Fine-tuning narrows but does not erase the gap

Under full fine-tuning rather than a frozen probe, INTENSITY reaches 0.8947 (mean-pool head) against RANDOM’s 0.8868, a gap of +0.0079 compared with +0.0109 frozen. Taking each arm’s best head instead, the gap is +0.0069. The pretraining difference is partially, but not entirely, absorbed by supervised adaptation — consistent with the masking policy shaping the representation rather than merely the optimisation start point. These runs used a different head and schedule from the frozen probes and are never pooled with them.

K Fine-tuned results

Table 7: Primary contrasts: matched epoch *and* matched precision. Δ is a paired difference on identical test cases; the interval is a 10,000-resample paired bootstrap and p is a DeLong test for correlated ROC curves. q is Benjamini–Hochberg over these nine pre-specified contrasts. Every comparison against the null excludes zero.

contrast	epoch	Δ AUC	95% CI	p	q
ENVELOPE – RANDOM	50	+0.0120	[+0.0068, +0.0173]	<0.0001	<0.0001
ENVELOPE – RANDOM	75	+0.0080	[+0.0028, +0.0132]	0.0025	0.0045
ENVELOPE – RANDOM	100	+0.0062	[+0.0010, +0.0113]	0.0199	0.0299
INTENSITY – RANDOM	50	+0.0099	[+0.0051, +0.0149]	<0.0001	0.0001
INTENSITY – RANDOM	75	+0.0113	[+0.0063, +0.0164]	<0.0001	<0.0001
INTENSITY – RANDOM	100	+0.0109	[+0.0057, +0.0160]	<0.0001	<0.0001
INTENSITY – ENVELOPE	50	-0.0020	[-0.0069, +0.0029]	0.4114	0.4114
INTENSITY – ENVELOPE	75	+0.0033	[-0.0013, +0.0079]	0.1491	0.1677
INTENSITY – ENVELOPE	100	+0.0047	[+0.0004, +0.0091]	0.0294	0.0378

Table 8: Full fine-tuning, same test split. Reported separately from frozen probes and never pooled with them.

policy	head	test AUC
INTENSITY	mean-pool	0.894668
INTENSITY	cross-attention	0.893717
INTENSITY	attentive	0.890100
RANDOM	attentive	0.887763
RANDOM	cross-attention	0.887178
RANDOM	mean-pool	0.886756

All masking arms on the SAME 5 slices (volume idx 2652) - one colour per predictor block, production samplers

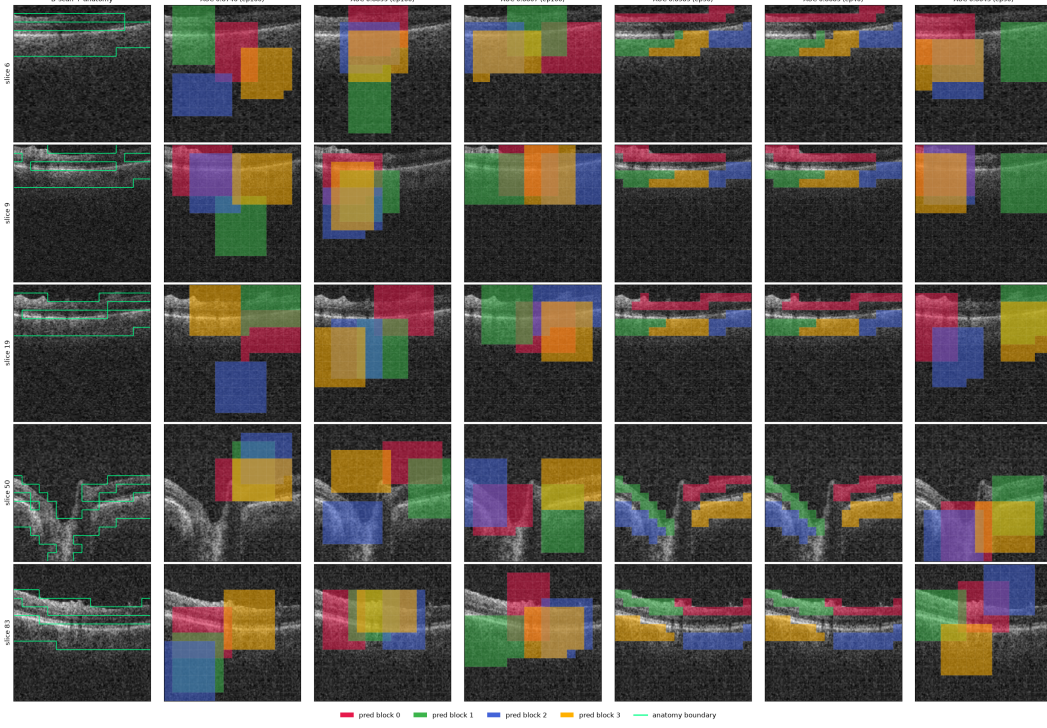


Figure 4: All six masking policies (columns) on five B-scans (rows) spanning one volume, drawn with the production samplers. Leftmost column is the unmasked B-scan. Each predictor target block has its own colour; the green contour is the anatomy reference. The random, envelope, and cover arms place axis-aligned rectangles; the anatomy arms grow cell-wise targets that follow the retinal band and therefore change shape from slice to slice.

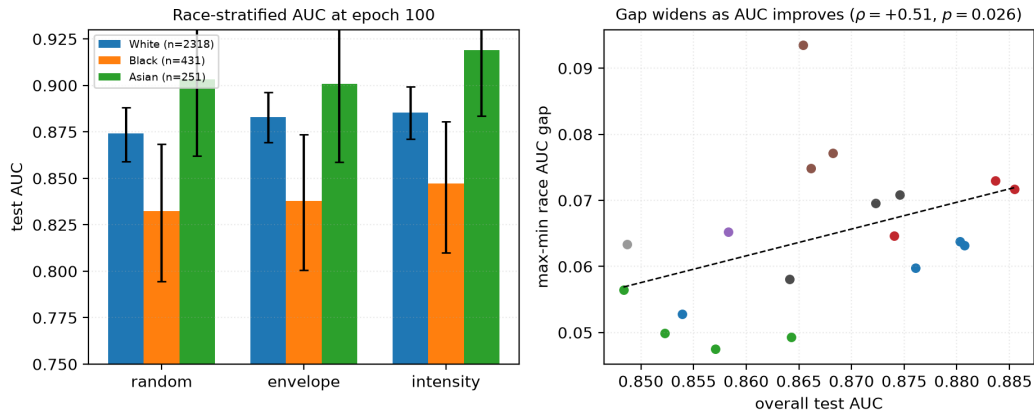


Figure 5: **Left:** race-stratified AUC at epoch 100 with bootstrap intervals. The black subgroup is lowest under every policy, but note that *every* group improves from RANDOM to INTENSITY. **Right:** max-min race gap against aggregate AUC across 19 checkpoints. The fitted slope is positive but does not survive multiplicity correction or branch-level aggregation (Table 4), so we draw no trend conclusion from it.

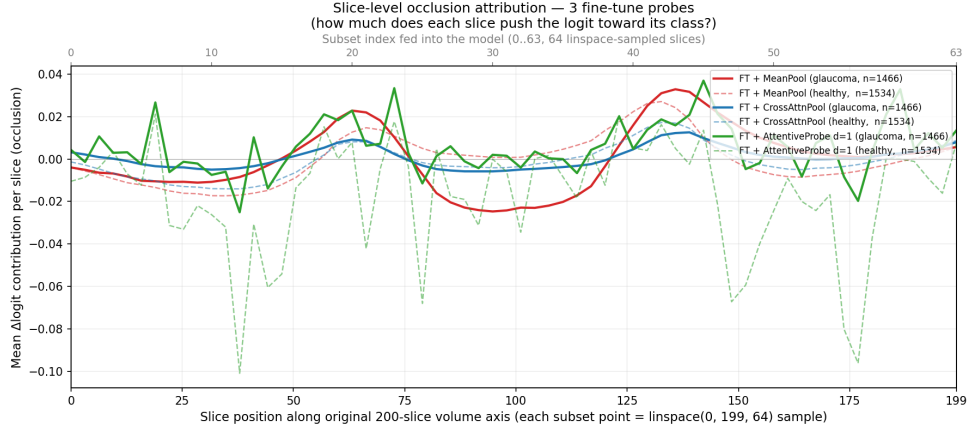


Figure 6: Single-slice occlusion attribution against native slice position, averaged over all 3000 test volumes, for all three fine-tuned probes. Despite spanning zero to 7.17M probe parameters they converge on the same structure (mean-pool vs cross-attention pool $r=0.94$). The y -axis is *signed* Δlogit : peaks are slices whose removal lowers the logit, the central dip is slices whose removal raises it, and only slices near zero are genuinely unused.

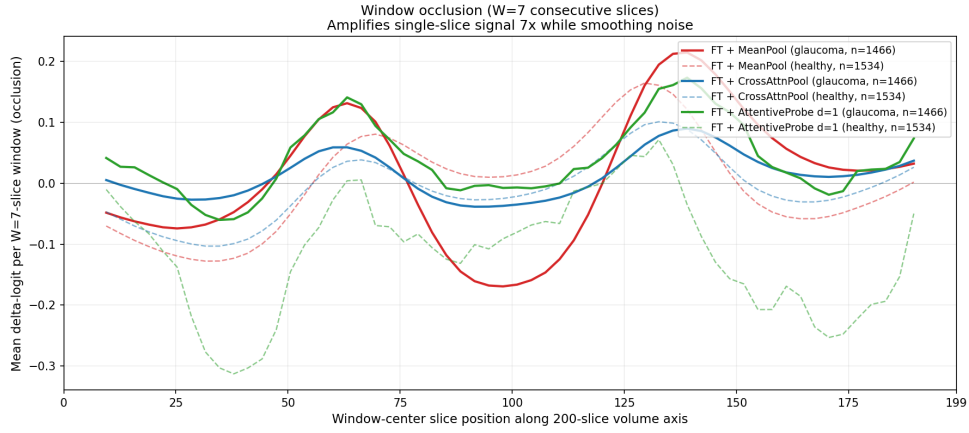


Figure 7: Window occlusion ($W=7$) amplifies the same signal roughly sevenfold and suppresses single-volume noise. For mean-pooled models, single-slice occlusion systematically underestimates attributed signal; the windowed variant recovers about $25\times$ more. We recommend it as the default primitive for this class of model.

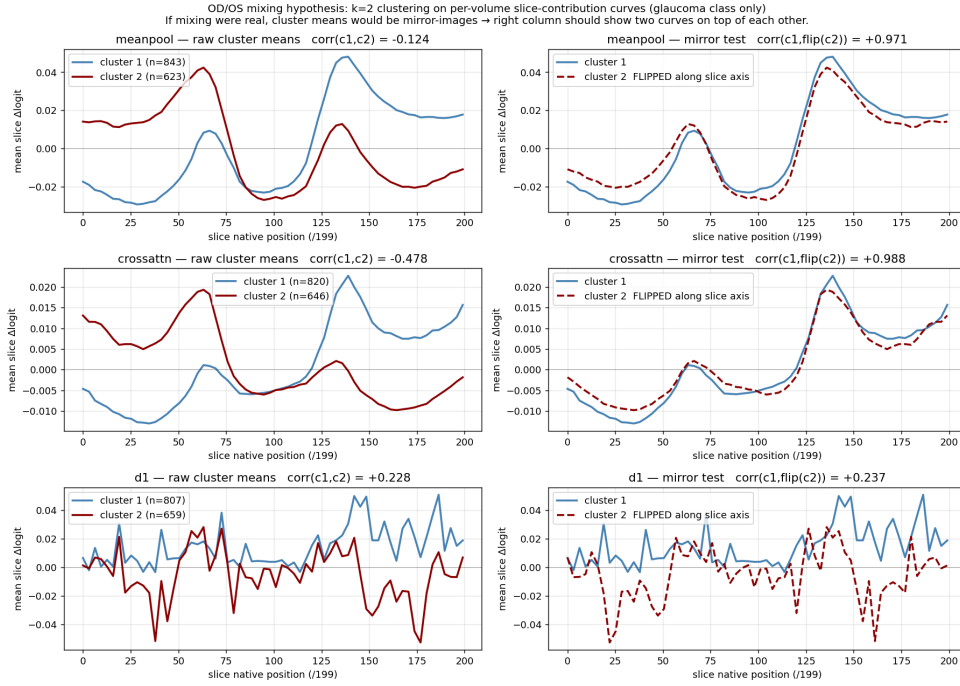


Figure 8: The two-peak structure is an OD/OS storage artefact. Clustering per-volume attribution curves into two groups returns near-perfect mirror images of one another, which is the direct signature of laterality flipping rather than bilateral anatomy.

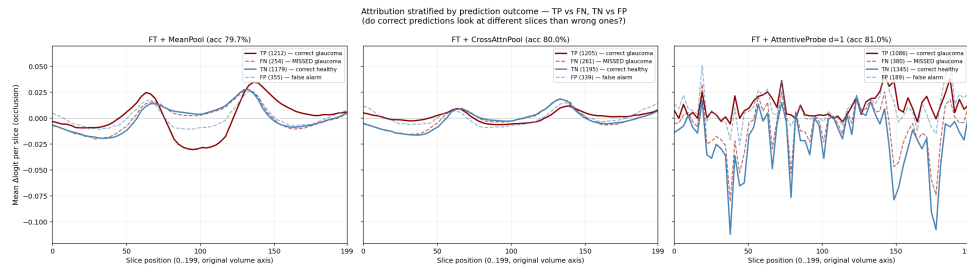


Figure 9: Attribution stratified by TP / FN / TN / FP. Errors reproduce the correct pattern at lower amplitude rather than selecting different anatomy.

Occlusion attribution — representative B-scans across the 3 fine-tune probes
Left pane of each row: original B-scan. Right pane: per-patch $\Delta\logit$ overlay.

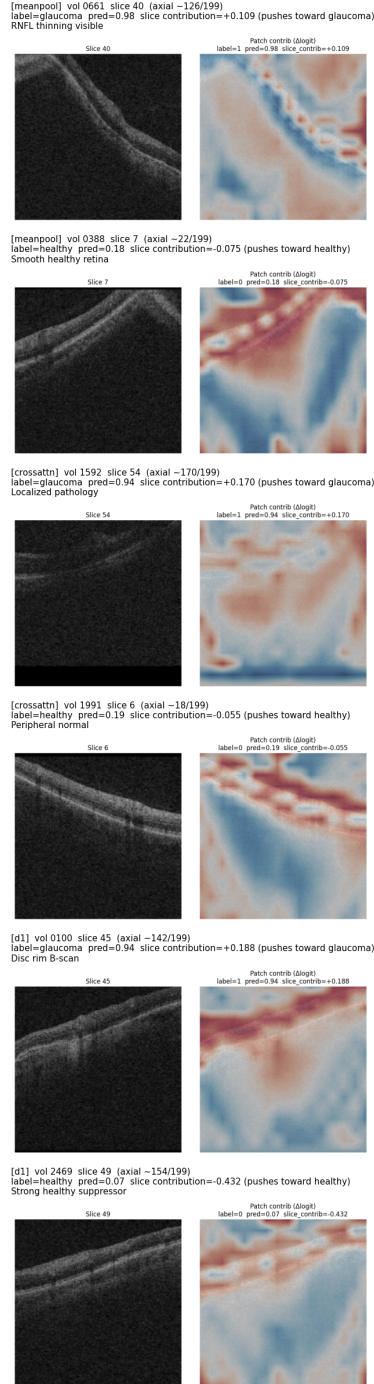


Figure 10: Per-patch occlusion heat maps on individual B-scans, drawn on a shared scale with the slice mean subtracted so that maps are comparable across volumes. Between 84% and 91% of patches have a 95% bootstrap interval excluding zero on the glaucoma class ($B=500$), so attribution is broadly distributed rather than concentrated in a few patches. Cross-probe agreement is strong at slice level ($r=0.94$) and moderate at patch level (per-volume $r \approx 0.35-0.48$).

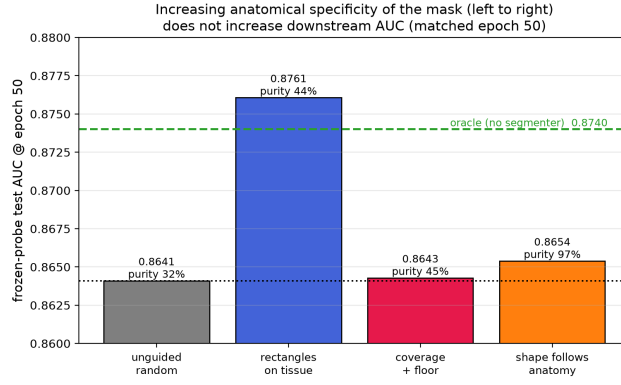


Figure 11: The matched-epoch-50 result arranged as a ladder of increasing anatomical specificity. Companion to Figure 3.

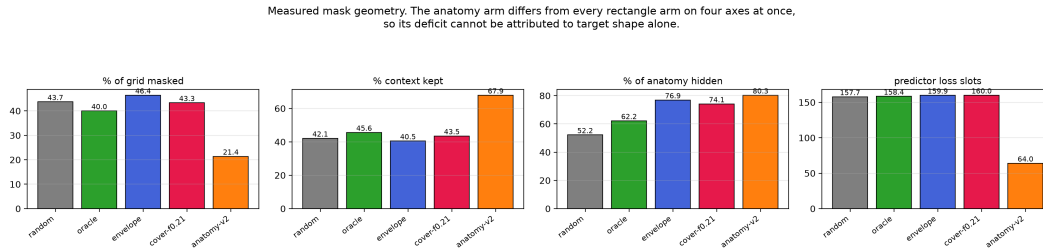


Figure 12: Measured mask geometry per policy, 600 held-out slices. The four rectangle policies are closely matched on every axis, so their comparison is near single-variable. The anatomy policy diverges on all four simultaneously — half the masking ratio, $1.6\times$ the context, $0.4\times$ the predictor loss slots — which is why we decline to attribute its deficit to target shape.